

Name: _____

Date: _____

DECIMALS, (ROUNDING DECIMALS TO DECIMAL PLACES)

LO: I can round decimal numbers to a given number of decimal places.

Rounding to decimal places

Rounding to decimal places (d.p.) means shortening a decimal to a given number of digits after the **decimal point**.

Rule: Look at the NEXT digit. If it is 5 or more, round up. If less than 5, round down (keep the same).

Examples

●	3.147 to 1 d.p. = 3.1	4 < 5, round down
●	3.147 to 2 d.p. = 3.15	7 ≥ 5, round up
●	6.998 to 2 d.p. = 7.00	rounds up to 7.00
●	3.147 to 1 d.p. = 3.2	look at 4, not 7
●	6.998 to 2 d.p. = 6.10	should be 7.00

Key idea

Look at the digit AFTER the one you want. 5 or above: round up. Below 5: round down.

3.147 to 2 d.p. look at 7 (third d.p.) 5 **3.15**

Critical digit decides

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - rounding to 1 d.p.

Round 4.63 to 1 decimal place.

We want 1 d.p.: keep one digit after the point

Look at the 2nd d.p.: 3

3 < 5, so round down: 4.6

= 4.6

The digit after is 3 (less than 5), so we keep 4.6.

Example 2 - rounding to 2 d.p.

Round 7.386 to 2 decimal places.

We want 2 d.p.: keep two digits after the point

Look at the 3rd d.p.: 6

6 ≥ 5, so round up: 7.39

= 7.39

The 6 forces the previous digit (8) to round up to 9.

Example 3 - rounding up carries

Round 2.997 to 2 decimal places.

Look at 3rd d.p.: 7 ≥ 5, round up

2.99 rounds up to 3.00

= 3.00

= 3.00

Rounding can carry through: 2.99 + 0.01 = 3.00.

Example 4 - to 3 d.p.

Round 0.12345 to 3 decimal places.

Want 3 d.p.: look at 4th digit: 4

4 < 5, round down

= 0.123

= 0.123

The fourth decimal digit (4) is less than 5.

YOUR TURN – PRACTICE (deliberate practice)

1. Round to 1 decimal place

Round each number to 1 d.p.

- a) 3.47
- b) 8.12
- c) 5.95
- d) 0.68

2. Round to 2 decimal places

Round each number to 2 d.p.

- a) 4.567
- b) 3.142
- c) 7.995
- d) 0.1234

3. Round to 3 decimal places

Round each number to 3 d.p.

- a) 2.71828
- b) 0.12345
- c) 9.99951
- d) 6.54321

4. Word problems

Round the answer appropriately.

- a) $10 \div 3 = 3.333...$ to 2 d.p.
- b) $\pi = 3.14159...$ to 3 d.p.
- c) £7 shared among 3 people (to nearest penny)
- d) $22 \div 7 = 3.14285...$ to 2 d.p.

5. Mixed rounding

Round to the number of d.p. stated.

- a) 6.4812 to 1 d.p.
- b) 3.0962 to 2 d.p.
- c) 0.999 to 1 d.p.
- d) 12.3456 to 3 d.p.
- e) 7.005 to 2 d.p.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 3.147 to 1 d.p. = 3.1 ☐

Correct answer: _____

Reason: _____

b) Look at the digit you want to keep, not the next one ☐

Correct answer: _____

Reason: _____

c) 6.998 rounded to 2 d.p. = 7.00 ☐

Correct answer: _____

Reason: _____

d) 5 always rounds down ☐

Correct answer: _____

Reason: _____

e) Rounding 4.95 to 1 d.p. gives 5.0 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A calculator shows $7 \div 6 = 1.1666666...$ Round this to 1, 2, and 3 decimal places. What pattern do you notice?

Expression: _____

Simplified: _____

b) A measurement is **3.45 cm** rounded to 2 d.p. What are the minimum and maximum values the actual measurement could be? (This is the error interval.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

